

Collisional Harvesting of Dissipatively Stabilized Spin Coherence

Why the swap-engine draft failed, the physical picture that survives,
and a closed companion model with impedance-matched power

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Abstract

A non-collinear quantum-dot spin valve produces less thermoelectric power than its collinear counterpart, while the same non-collinearity generates spin coherence in the dot. An earlier draft proposed to restore collinear performance and charge an ancilla battery by swapping the dot onto a tape of qubits. That construction fails for physical reasons catalogued in Sec. 3. The idea that survives is different: export a transport-generated coherent resource into a battery, treating electrical power and battery ergotropy as two coexisting outputs rather than as a restoration protocol.

We reformulate the engine with charge- and energy-conserving exchange collisions and a stroboscopic fixed point. To isolate the harvesting physics, we solve a minimal companion model—a V-type three-level system whose bath-stabilized doublet coherence is repeatedly drained onto ground-state ancillas. With a finite matched gap that makes the collision work-free and the ancilla energetically nontrivial, the collision map, stroboscopic fixed point, exact qubit ergotropy, and charging power are obtained in closed form. The central result is an impedance-matching law

$$\Gamma_{\text{ext}} = r(1 - \cos \theta) \simeq \gamma_C,$$

which yields a small-angle harvesting bound $P_{\text{erg}}^{\text{max}} \sim \gamma_C \Delta c_0^2 / (2|z_A|)$ and a flat plateau in the exchange angle. Finite precession, directed dark-state renewal, coupling-strength realizability, and the operational accessibility of coherent ergotropy are treated explicitly. The companion model is ready as an analytic foundation; the full spin-valve engine is the natural next layer.

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1 Executive status

The project splits cleanly into two layers.

1. **Companion model (this note, closed analytically).** A V-type system with bath-stabilized doublet coherence, repeatedly coupled to a stream of spin ancillas. Collision map, fixed point, ergotropy, and optimal power are available in closed form.
2. **Full engine (next).** A non-collinear spin-valve transport device with a collision tape. This should be added only after the companion model is physically closed and the figures of merit are settled.

The companion-model result that organizes the whole harvesting problem is the impedance-matching condition

$$\Gamma_{\text{ext}} = r(1 - \cos \theta) \simeq \gamma_C, \quad (1)$$

where r is the collision rate, $\theta = g\tau$ is the exchange angle, and γ_C is the coherence regeneration rate. In the simplest zero-phase, weak-extraction limit,

$$P_{\text{erg}}^{\text{max}} \sim \frac{\gamma_C \Delta c_0^2}{2|z_A|}, \quad (2)$$

with c_0 the free stabilized coherence and Δ the matched ancilla gap. The rate is linear in bath regrowth and quadratic in the stabilized coherence.

Main physical repairs relative to earlier drafts.

1. Exact degeneracy plus nonzero ancilla ergotropy is impossible under strict energy conservation; a finite matched splitting $\Delta_A = \Delta > 0$ is required.
2. Symmetric bright–dark mixing equilibrates the doublet and kills the steady coherence; directed dark leakage (or phase-locked reset) is needed.
3. Finite-coupling realizability gives $\theta_{\min} \sim 2\gamma_C/g$ (linear in γ_C/g), not a square-root constraint.
4. Two ancillas do *not* generically unlock single-copy coherent ergotropy; accessible work must be computed with an explicit discharge model or phase reference.

2 The original proposal

2.1 Model

A magnetic quantum dot with two non-degenerate levels,

$$H_D = \epsilon_d d_{\downarrow}^{\dagger} d_{\downarrow} + \epsilon_u d_{\uparrow}^{\dagger} d_{\uparrow}, \quad \Delta \equiv \epsilon_d - \epsilon_u > 0, \quad (3)$$

is tunnel-coupled to magnetic leads $\alpha \in \{L, R\}$ at temperatures $T_L > T_R$ and chemical potentials μ_L, μ_R . The left lead is polarized along the dot axis; the right lead is tilted by (θ_R, ϕ_R) . Non-collinearity enters through the broadening matrices

$$\Gamma_{\alpha} = \gamma \begin{pmatrix} 1 + p \cos \theta_{\alpha} & p e^{-i\phi_{\alpha}} \sin \theta_{\alpha} \\ p e^{i\phi_{\alpha}} \sin \theta_{\alpha} & 1 - p \cos \theta_{\alpha} \end{pmatrix}, \quad \Sigma_{\alpha} = -\frac{i}{2} \Gamma_{\alpha}. \quad (4)$$

A tape of ancilla qubits with gap $\hbar\omega_a$, each prepared in $\rho_a = \text{diag}(1 - \alpha, \alpha)$, moves past the dot. The proposed cycle: the dot equilibrates with the right bath into a coherent state ρ_R ; a SWAP exchanges dot and ancilla states; the dot then couples to the left bath; repeat. Transport was computed from stationary NEGF,

$$I = \frac{q}{h} \int dE (f_L - f_R) \text{Tr} [\Gamma_L G \Gamma_R G^{\dagger}], \quad G = (E\mathbb{K} - H_D - \Sigma_L - \Sigma_R)^{-1}, \quad (5)$$

with $P = IV_{\text{app}}$, $\eta = P/\dot{Q}_h$, and the dot “state” identified with $\rho = \int dE G \Sigma_{\text{in}} G^\dagger$, $\Sigma_{\text{in}} = \Gamma_L f_L + \Gamma_R f_R$. The swap energetics were quantified by mean-energy differences

$$\mathcal{W}_d = \text{Tr}[H_d(\rho_a - \rho_R)], \quad \mathcal{W}_a = \text{Tr}[H_a(\rho_R - \rho_a)], \quad \mathcal{W}_U = -(\mathcal{W}_d + \mathcal{W}_a), \quad (6)$$

and the claimed results were (i) the swap makes the engine “effectively collinear,” recovering the collinear power–efficiency loop, and (ii) the ancilla battery charges at no drive cost when $\hbar\omega_a = \Delta$.

3 Why the old work did not work

3.1 A stationary formalism cannot register a stroke protocol

Equation (5) describes the unique nonequilibrium steady state of a dot coupled to *both* leads continuously; it is fixed by $\{H_D, \Gamma_{L,R}, f_{L,R}\}$ alone. Any state deposited in the dot by a swap relaxes back to this same steady state on the timescale $\hbar/\gamma$, after which the currents are unchanged. The old comparison of collinear and non-collinear power–efficiency loops compared two different Γ_R matrices; the swap never entered the transport calculation, so the enhancement claim was never actually computed. For the ancilla stream to affect transport at all, the swap must appear *in the equation of motion*—as a stroboscopic collision map or a periodically driven master equation—and the currents must be evaluated at the resulting *periodic* stationary state.

3.2 Mean energy is not ergotropy

The quantities in Eq. (6) are energy changes. Extractable work is ergotropy,

$$\mathcal{W}(\rho, H) = \text{Tr}(\rho H) - \text{Tr}(\pi_\rho H), \quad (7)$$

with π_ρ the passive state. For a coherent ρ_R the difference between the two is precisely the coherent contribution [1]—the very resource at stake—so the old “battery charging” plots measured the wrong quantity. Worse, coherent ergotropy is frame-dependent: without a phase reference, part of it is inaccessible to energy-conserving discharge operations [2]. Any redo must report total, coherent, and *operationally accessible* ergotropy of the outgoing ancilla.

3.3 The dot is not a qubit: Fock space and superselection

$\rho = \int dE G \Sigma_{\text{in}} G^\dagger$ is the single-particle correlation matrix $\langle d_o^\dagger d_{o'} \rangle$, not a normalized two-level state. The dot Fock space contains $\{|0\rangle, |\uparrow\rangle, |\downarrow\rangle, |\uparrow\downarrow\rangle\}$; only the spin coherence *within the singly occupied sector* is a qubit-like degree of freedom. A SWAP between the full dot and a spin qubit moves weight between charge sectors and violates particle-number superselection: no charge-conserving interaction can implement it. The exchange must be conditioned on single occupancy, which introduces an unavoidable occupation penalty $P_1 = P_\uparrow + P_\downarrow$ on any charging rate.

3.4 Incompatible timescales and couplings

The draft simultaneously assumed broadening-dominated transport (strong lead coupling), weak coupling in the figure parameters, an instantaneous swap, and an ancilla transit “short compared to relaxation.” These cannot coexist. The physical control parameters are the exchange angle $g\tau$ and the ratios of collision, relaxation, and waiting times $(\tau, \hbar/\gamma, T)$; all must appear explicitly in the stationary map.

3.5 Coherence was never shown to cause the power loss

The implicit causal chain was *non-collinearity* $\rightarrow$ *dot coherence* $\rightarrow$ *reduced power*. Only the first arrow is established. Non-collinearity suppresses the current *geometrically*—transmission mismatch between lead polarization axes (spin-valve filtering)—and this suppression persists even if the local dot coherence is erased, because Γ_R in Eq. (4) stays non-collinear. Erasing or exporting the coherence therefore need not recover the collinear loop. The decisive test is *population-preserving dephasing* restricted to the spin sector: only if dephasing alone raises P_{el} is the coherence itself detrimental. If not, the honest physical statement is *resource partitioning*: non-collinear transport generates a coherent by-product that coexists with geometric losses, and the battery is an additional output channel—not a repair of the engine.

3.6 The parameter regime destroys the resource it needs

The legacy parameters give $\Delta/\gamma \approx 2000$. For such well-split levels, lead-induced coherence in the energy basis is parametrically suppressed, and a secular weak-coupling master equation decouples it from populations entirely. In the regime where the old calculation lived, there is essentially nothing to harvest. Retaining the resource requires (i) operating near degeneracy, $\Delta \lesssim \gamma$, and (ii) a nonsecular generator (Redfield / first-order von Neumann / singular-coupling near degeneracy) whose validity and positivity are checked.

3.7 A full swap does not isolate coherence

An iSWAP transfers populations along with coherence. A battery-ergotropy gain, or a current change, can then be dominated by ordinary population resetting and misattributed to coherence harvesting. The clean theoretical control is a *matched-population ancilla*: incoming ancillas carry the dot’s instantaneous diagonal populations and zero coherence, so the collision transfers coherence with no population gradient. Comparing (no intervention / dephasing / classical population reset / incoherent exchange / coherent partial swap / matched-population swap / collinear reference) is what attributes the effect; the swap angle $g\tau$ should be scanned continuously rather than fixed at the full-swap point.

4 The physical picture that survives

4.1 The resource is a misaligned spin vector

The stationary coherence $\rho_{\uparrow\downarrow}$ is basis-dependent: it is equivalent to a stationary dot spin polarization $\vec{S}$ tilted away from the quantization axis of H_D , maintained by the competition between the two lead polarizations and precessing in the lead-induced exchange field. A local basis rotation converts it to populations at zero cost, which is exactly why the extractable resource must be defined relative to the *battery’s* energy eigenbasis and the allowed number- and energy-conserving operations [2]. What the ancilla harvests is the component of $\vec{S}$ transverse to the ancilla quantization axis. The frame-aware figures of merit are the misalignment angle $\vartheta = \angle(\vec{S}, \hat{z}_A)$ and transverse polarization $|\vec{S}| \sin \vartheta$; “coherence” is shorthand.

4.2 Near-degeneracy is the operating point

Three independent requirements select the same regime, $\Delta \lesssim \gamma$:

1. stationary energy-basis coherence is $O(1)$ instead of $O(\gamma/\Delta)$, and nonsecular terms survive;

2. the degenerate-subspace exchange (below) can be made to commute with the free Hamiltonian after a finite matched-gap repair, making the collision work-free;
3. a near-degenerate doublet coupled dissipatively supports long-lived coherence—the same mechanism as in V-type three-level systems [3]—so the bath continuously regrows the resource between collisions.

The near-degenerate spin valve is the fermionic-transport analogue of the V-system, and the legacy choice $\Delta = 0.1 k_B T_L \gg \gamma$ was simply the wrong regime.

5 The reformulated engine

5.1 State space and interactions

Take a strong-Coulomb-blockade transport dot, $\mathcal{H}_D = \text{span}\{|0\rangle, |\uparrow\rangle, |\downarrow\rangle\}$,

$$H_D = \varepsilon (n_\uparrow + n_\downarrow) + \frac{\Delta}{2} (n_\uparrow - n_\downarrow), \quad (8)$$

and a second, permanently singly occupied (Coulomb-blockaded) dot as the ancilla, $H_A = \frac{\Delta_A}{2} \tau_z$. The exchange acts only in the singly occupied sector of the transport dot,

$$H_{\text{int}} = \hbar g (|\uparrow_D \downarrow_A\rangle \langle \downarrow_D \uparrow_A| + \text{h.c.}), \quad (9)$$

which conserves particle number in each dot, $[H_{\text{int}}, N_D + N_A] = 0$, is trivial on $|0\rangle_D$, and at resonance $\Delta_A = \Delta$ is energy-conserving on the active block (Sec. 7). Acting for a time τ it generates a partial iSWAP of angle $g\tau$ on the one-magnon subspace.

5.2 Dynamics and stationary state

Between collisions the dot evolves under a nonsecular lead generator $\mathcal{L}_{\text{leads}}$ that retains spin coherences (Redfield or first-order von Neumann; positivity monitored). With collision period T and fresh ancillas $\rho_{A,\text{in}}$, the stroboscopic map is

$$\rho_D^{(n+1)} = \text{Tr}_A \left\{ U_\tau \left[e^{\mathcal{L}_{\text{leads}}(T-\tau)} \rho_D^{(n)} \otimes \rho_{A,\text{in}} \right] U_\tau^\dagger \right\}, \quad \rho_D^* = \Phi(\rho_D^*), \quad (10)$$

and *every* reported quantity—charge current, lead energy currents, hot-bath heat current, outgoing ancilla state—is evaluated at the periodic fixed point ρ_D^* . (In a refined treatment the lead dissipator acts during the collision as well.)

5.3 Bookkeeping

Ergotropy is a work capacity carried inside the outgoing ancilla energy, not an independent energy current; it must not be added to a first-law balance. The per-period balance at the fixed point is

$$0 = \sum_\alpha \bar{J}_{E,\alpha} + P_{\text{switch}} - r \Delta E_A, \quad r = 1/T, \quad (11)$$

with $P_{\text{el}} = -(\mu_L - \mu_R) \bar{J}_N$ and the charging rate $P_{\text{erg}} = r \Delta \mathcal{W}_A$ reported separately, split into coherent and incoherent parts [1] and into accessible/inaccessible parts [2]. The ancilla tape is a low-entropy resource; its preparation cost is bounded below by $\Delta F_A = \Delta E_A - T_0 \Delta S_A$ against a declared ambient temperature T_0 , and the conservative net figure of merit is

$$P_{\text{useful}} = P_{\text{el}} + r \mathcal{W}_{\text{accessible}} - P_{\text{switch}} - P_{\text{prep}}. \quad (12)$$

5.4 Central physical question

The question the model answers is no longer “does the swap restore collinear performance” but: *how does a non-collinear transport steady state partition its nonequilibrium free energy between electrical output and exported battery ergotropy?* The primary deliverable is the Pareto frontier $(P_{\text{el}}, P_{\text{erg}})$ as a function of $(g\tau, \gamma T, \theta_R, p, \Delta/\gamma)$, together with the dephasing control of Sec. 3 that decides whether any regime exists where exporting coherence *also* raises the electrical output. Known no-go results for minimal qubit repeated-interaction engines [5] do not directly apply here (fermionic particle flow, chemical-potential bias, charge sectors, battery tape), but they constrain the limits the model must recover when those features are switched off.

6 Companion model: definitions

The harvesting physics can be isolated from transport complications in a V-type three-level system whose bath-stabilized doublet coherence is drained onto a stream of ancillas [3]. This section fixes notation; the following sections give the full analytic development.

6.1 System variables

The companion system has ground state $|0\rangle$ and doublet states $|1\rangle, |2\rangle$. In the transport language these play the role of the singly occupied spin states. Define

$$s = \rho_{11} + \rho_{22}, \quad d = \rho_{11} - \rho_{22}, \quad C = \rho_{12}. \quad (13)$$

Here s is the doublet weight, d is the population imbalance in the energy basis, and C is the coherence to be harvested.

The bright and dark states are

$$|B\rangle = \frac{|1\rangle + |2\rangle}{\sqrt{2}}, \quad |D\rangle = \frac{|1\rangle - |2\rangle}{\sqrt{2}}. \quad (14)$$

If C is real,

$$p_B = \frac{s}{2} + C, \quad p_D = \frac{s}{2} - C, \quad p_B - p_D = 2C. \quad (15)$$

Thus “coherence” is equivalently a bright–dark population imbalance.

6.2 Ancilla

Each ancilla is a qubit with input state

$$\rho_A^{\text{in}} = (1 - \alpha)|\downarrow\rangle\langle\downarrow| + \alpha|\uparrow\rangle\langle\uparrow|. \quad (16)$$

The physically relevant operating point for local ancilla ergotropy is usually $\alpha = 0$ (ground-state ancillas). The matched-population point $\alpha = 1/2$ is a diagnostic control: it removes system coherence without locally charging the ancilla.

7 Finite-gap, work-free resonant repair

7.1 Why exact degeneracy alone is not enough

Take a degenerate doublet,

$$H_S = \omega(|1\rangle\langle 1| + |2\rangle\langle 2|), \quad (17)$$

and an ancilla Hamiltonian

$$H_A = \omega_a |\uparrow\rangle\langle\uparrow|_A. \quad (18)$$

The exchange interaction is

$$H_{\text{int}} = \hbar g (|1 \downarrow_A\rangle\langle 2 \uparrow_A| + |2 \uparrow_A\rangle\langle 1 \downarrow_A|). \quad (19)$$

The two coupled states have free energies

$$E_{1\downarrow} = \omega, \quad E_{2\uparrow} = \omega + \omega_a. \quad (20)$$

Therefore

$$[H_S + H_A, H_{\text{int}}] = 0 \iff \omega_a = 0. \quad (21)$$

But if $\omega_a = 0$, the ancilla Hamiltonian is proportional to the identity and the ancilla carries no energetic ergotropy. Hence exact degeneracy, strict work-freeness, and nonzero ancilla ergotropy are mutually incompatible.

7.2 Finite matched splitting

Introduce a finite splitting in the doublet,

$$H_S = \omega (|1\rangle\langle 1| + |2\rangle\langle 2|) + \frac{\Delta}{2} (|1\rangle\langle 1| - |2\rangle\langle 2|), \quad (22)$$

and match the ancilla gap:

$$H_A = \Delta |\uparrow\rangle\langle\uparrow|_A. \quad (23)$$

Then the active states have identical total free energy:

$$E_{1\downarrow} = \omega + \frac{\Delta}{2}, \quad E_{2\uparrow} = \omega - \frac{\Delta}{2} + \Delta = \omega + \frac{\Delta}{2}. \quad (24)$$

Therefore, on the active block,

$$H_S + H_A = \left(\omega + \frac{\Delta}{2}\right) \mathbb{I}_{2 \times 2}, \quad (25)$$

and

$$[H_S + H_A, H_{\text{int}}] = 0. \quad (26)$$

The collision is strictly energy-conserving while the ancilla gap is nonzero. Thus the collision can be work-free and the outgoing ancilla can carry nonzero ergotropy.

The same interaction preserves particle number in the intended exchange sector:

$$[H_{\text{int}}, N_S + N_A] = 0, \quad (27)$$

and H_{int} is trivial outside the allowed singly occupied block.

8 Collision map in closed form

8.1 Block unitary

Let $\theta = g\tau$. On the active subspace $\{|1 \downarrow_A\rangle, |2 \uparrow_A\rangle\}$,

$$U_\tau = e^{-iH_{\text{int}}\tau/\hbar} = \begin{pmatrix} \cos \theta & -i \sin \theta \\ -i \sin \theta & \cos \theta \end{pmatrix}. \quad (28)$$

Equivalently,

$$U|1 \downarrow\rangle = \cos \theta |1 \downarrow\rangle - i \sin \theta |2 \uparrow\rangle, \quad (29)$$

$$U|2 \uparrow\rangle = \cos \theta |2 \uparrow\rangle - i \sin \theta |1 \downarrow\rangle. \quad (30)$$

All other basis states are spectators.

8.2 System reduced map

The input is $\rho \otimes \rho_A^{\text{in}}$. Since ρ_A^{in} is diagonal, the calculation separates into two classical branches.

Branch $\downarrow_A$, weight $1 - \alpha$. The active state is $|1 \downarrow_A\rangle$; $|2 \downarrow_A\rangle$ is a spectator. After conjugation by U and tracing over the ancilla,

$$\rho_{11} \mapsto \cos^2 \theta \rho_{11}, \quad (31)$$

$$\rho_{22} \mapsto \rho_{22} + \sin^2 \theta \rho_{11}, \quad (32)$$

$$C \mapsto \cos \theta C. \quad (33)$$

Branch $\uparrow_A$, weight α . The active state is $|2 \uparrow_A\rangle$; $|1 \uparrow_A\rangle$ is a spectator:

$$\rho_{22} \mapsto \cos^2 \theta \rho_{22}, \quad (34)$$

$$\rho_{11} \mapsto \rho_{11} + \sin^2 \theta \rho_{22}, \quad (35)$$

$$C \mapsto \cos \theta C. \quad (36)$$

Weighted sum. Combining both branches and using

$$(1 - \alpha)\rho_{11} - \alpha\rho_{22} = \frac{1}{2}[s(1 - 2\alpha) + d], \quad (37)$$

we obtain

$$s' = s, \quad (38)$$

$$d' = \cos^2 \theta d - \sin^2 \theta (1 - 2\alpha)s, \quad (39)$$

$$C' = \cos \theta C. \quad (40)$$

The invariant $s' = s$ is superselection in miniature: the exchange never moves weight between the doublet and the ground sector.

8.3 Ancilla reduced map

Tracing over the system gives

$$p_{\uparrow}^{\text{out}} = \alpha + \sin^2 \theta [(1 - \alpha)\rho_{11} - \alpha\rho_{22}] = \alpha + \frac{\sin^2 \theta}{2}[s(1 - 2\alpha) + d], \quad (41)$$

and

$$\rho_{\uparrow\downarrow}^{A,\text{out}} = -i \sin \theta (1 - 2\alpha)C. \quad (42)$$

The Bloch components of the outgoing ancilla are

$$x_A = 2 \operatorname{Re} \rho_{\uparrow\downarrow}^{A,\text{out}}, \quad (43)$$

$$y_A = -2 \operatorname{Im} \rho_{\uparrow\downarrow}^{A,\text{out}}, \quad (44)$$

$$z_A = 2p_{\uparrow}^{\text{out}} - 1. \quad (45)$$

For real C this reduces to

$$x_A = 0, \quad y_A = 2 \sin \theta (1 - 2\alpha)C, \quad z_A = -(1 - 2\alpha) + \sin^2 \theta [s(1 - 2\alpha) + d]. \quad (46)$$

8.4 The $\alpha = 1/2$ null result

At $\alpha = 1/2$,

$$\rho_{\uparrow\downarrow}^{A,\text{out}} = 0 \quad (47)$$

for every input C . Nevertheless, the system coherence is still reduced: $C \mapsto \cos \theta C$. The missing amplitude lives in system–ancilla correlations, proportional to

$$\frac{i \sin \theta}{2} C(|1\rangle\langle 1| - |2\rangle\langle 2|) \otimes |\uparrow\rangle\langle\downarrow|_A + \text{h.c.} \quad (48)$$

Thus $\alpha = 1/2$ is a clean *back-action control*: it removes system coherence without locally charging the ancilla.

9 Stroboscopic fixed point with finite gap

9.1 Phenomenological reset map

During the waiting time $\tau_c = T - \tau$ (or, in the short-collision idealization, $\tau_c = 1/r$), finite splitting produces free precession

$$C \mapsto e^{-i\Delta\tau_c} C. \quad (49)$$

Let

$$\phi = \Delta\tau_c, \quad q_C = e^{-\gamma_C\tau_c}. \quad (50)$$

In the simplest phase-locked phenomenological model,

$$C_{n+1} = q_C e^{-i\phi} \cos \theta C_n + (1 - q_C) c_0. \quad (51)$$

The fixed point is

$$C^* = \frac{(1 - q_C) c_0}{1 - q_C e^{-i\phi} \cos \theta}, \quad (52)$$

with magnitude

$$|C^*| = \frac{(1 - q_C) |c_0|}{\sqrt{1 + q_C^2 \cos^2 \theta - 2q_C \cos \theta \cos \phi}}. \quad (53)$$

Population variables do not acquire a Hamiltonian phase. With a single population relaxation factor $q = e^{-\gamma_d \tau_c}$ and target $d_{\text{ss}} = 0$,

$$s^* = s_{\text{ss}}, \quad d^* = -\frac{q \sin^2 \theta (1 - 2\alpha) s_{\text{ss}}}{1 - q \cos^2 \theta}. \quad (54)$$

Physical caveat on phase locking. A time-independent thermal bath for a non-degenerate doublet does not generically stabilize a lab-frame coherence c_0 . A finite-gap coherent target is a phase-locked, driven, rotating-frame, or near-degenerate nonsecular construction. This is acceptable as a companion model, but the paper must state which physical mechanism fixes the phase. If strict autonomy with a single time-independent thermal bath is required, the safest regime is $\Delta \ll \gamma_C$, with ergotropy proportional to the small but finite matched gap.

10 A physical bright/dark bath

10.1 Symmetric bright–dark mixing kills the resource

A proposed minimal bath was

$$\dot{p}_B = \Gamma_{\uparrow} p_0 - \Gamma_{\downarrow} p_B - \kappa(p_B - p_D), \quad (55)$$

$$\dot{p}_D = \kappa(p_B - p_D), \quad (56)$$

$$p_0 = 1 - p_B - p_D. \quad (57)$$

At steady state with $\kappa > 0$, the second equation forces $p_B = p_D$, hence $C = (p_B - p_D)/2 = 0$. Symmetric bright–dark mixing does not regenerate coherence; it equilibrates the bright and dark populations and destroys the resource.

Conclusion. Symmetric bright–dark mixing is a decoherence/noise control, not a regeneration bath. A renewable resource requires directed dark leakage or preferential bright-state repumping.

10.2 Directed dark leakage model

A minimal physically safer model is

$$\dot{p}_B = \Gamma_{\uparrow} p_0 - \Gamma_{\downarrow} p_B, \quad (58)$$

$$\dot{p}_D = -\kappa_D p_D, \quad (59)$$

$$p_0 = 1 - p_B - p_D. \quad (60)$$

Here the bath pumps and relaxes the bright state, while dark population leaks back to the reservoir/ground sector at rate κ_D . In vector form $v = (p_B, p_D)^T$,

$$\dot{v} = Mv + b, \quad M = \begin{pmatrix} -(\Gamma_{\uparrow} + \Gamma_{\downarrow}) & -\Gamma_{\uparrow} \\ 0 & -\kappa_D \end{pmatrix}, \quad b = \begin{pmatrix} \Gamma_{\uparrow} \\ 0 \end{pmatrix}. \quad (61)$$

The steady state is

$$p_D^{\text{ss}} = 0, \quad p_B^{\text{ss}} = \frac{\Gamma_{\uparrow}}{\Gamma_{\uparrow} + \Gamma_{\downarrow}} \equiv s_{\text{ss}}. \quad (62)$$

If detailed balance is imposed, $\Gamma_{\downarrow}/\Gamma_{\uparrow} = e^{\beta\omega}$, then

$$s_{\text{ss}} = \frac{1}{1 + e^{\beta\omega}}, \quad c_0 = \frac{s_{\text{ss}}}{2}. \quad (63)$$

10.3 Exact waiting-time map for directed leakage

Let

$$A = \Gamma_{\uparrow} + \Gamma_{\downarrow}, \quad q_B = e^{-A\tau_c}, \quad q_D = e^{-\kappa_D\tau_c}. \quad (64)$$

Starting from post-collision values (B_+, D_+) , the waiting evolution gives

$$D_- = q_D D_+, \quad (65)$$

and, for $A \neq \kappa_D$,

$$B_- = q_B B_+ + s_{\text{ss}}(1 - q_B) - \Gamma_{\uparrow} D_+ \frac{q_D - q_B}{A - \kappa_D}. \quad (66)$$

(The resonant case $A = \kappa_D$ is the limit $(q_D - q_B)/(A - \kappa_D) \rightarrow \tau_c e^{-A\tau_c}$.)

The collision in bright/dark population variables is

$$\begin{pmatrix} B_+ \\ D_+ \end{pmatrix} = R_\theta \begin{pmatrix} B_- \\ D_- \end{pmatrix}, \quad R_\theta = \frac{1}{2} \begin{pmatrix} 1 + \cos \theta & 1 - \cos \theta \\ 1 - \cos \theta & 1 + \cos \theta \end{pmatrix}. \quad (67)$$

Thus the full one-cycle map is affine,

$$v_{n+1} = \mathcal{B} R_\theta v_n + b_{\mathcal{B}}, \quad (68)$$

with

$$\mathcal{B} = \begin{pmatrix} q_B & -\Gamma_\uparrow \frac{q_D - q_B}{A - \kappa_D} \\ 0 & q_D \end{pmatrix}, \quad b_{\mathcal{B}} = \begin{pmatrix} s_{ss}(1 - q_B) \\ 0 \end{pmatrix}. \quad (69)$$

$$v^* = (\mathcal{K} - \mathcal{B} R_\theta)^{-1} b_{\mathcal{B}}. \quad (70)$$

This closed two-dimensional fixed point is the recommended replacement for the symmetric-mixing model.

11 Qubit ergotropy of the outgoing ancilla

11.1 General formula

For $H_A = \Delta |\uparrow\rangle\langle\uparrow|$, write the output qubit in Bloch form,

$$\rho_A = \frac{1}{2}(\mathcal{K} + x_A \sigma_x + y_A \sigma_y + z_A \sigma_z), \quad (71)$$

where $z_A = 2p_\uparrow - 1$ and $|r_A| = \sqrt{x_A^2 + y_A^2 + z_A^2}$. The energy is

$$E_A = \Delta p_\uparrow = \frac{\Delta}{2}(1 + z_A). \quad (72)$$

The passive state has excited population $(1 - |r_A|)/2$, so

$$E_A^{\text{pass}} = \frac{\Delta}{2}(1 - |r_A|). \quad (73)$$

$$\mathcal{W}_A = E_A - E_A^{\text{pass}} = \frac{\Delta}{2}(z_A + |r_A|). \quad (74)$$

If $z_A < 0$, the population state is passive and the ergotropy is purely coherent:

$$\mathcal{W}_A = \mathcal{W}_{A,\text{coh}} = \frac{\Delta}{2}(|r_A| - |z_A|). \quad (75)$$

11.2 Ground-state ancilla, $\alpha = 0$

For $\alpha = 0$,

$$z_A = -1 + \sin^2 \theta (s^* + d^*), \quad |\rho_{\uparrow\downarrow}^{A,\text{out}}| = \sin \theta |C^*|, \quad (76)$$

so

$$|r_A| = \sqrt{z_A^2 + 4 \sin^2 \theta |C^*|^2}, \quad (77)$$

and

$$\Delta \mathcal{W}_A = \frac{\Delta}{2} \left[z_A + \sqrt{z_A^2 + 4 \sin^2 \theta |C^*|^2} \right]. \quad (78)$$

For weak extraction, $\sin \theta |C^*| \ll |z_A|$,

$$\Delta \mathcal{W}_A \simeq \frac{\Delta \sin^2 \theta |C^*|^2}{|z_A|}. \quad (79)$$

12 Charging power and optimum: zero precession phase

12.1 Power function

Set $\phi = 0$ and use a single coherence rate $\gamma_C = \gamma$. Let

$$x = \gamma\tau_c = \frac{\gamma}{r}, \quad q = e^{-x}, \quad c = \cos\theta. \quad (80)$$

Then

$$\frac{C^*}{c_0} = \frac{1 - e^{-x}}{1 - ce^{-x}}. \quad (81)$$

In the weak-extraction approximation,

$$P_{\text{erg}}(r) = r\Delta\mathcal{W}_A = \frac{\gamma\Delta\sin^2\theta c_0^2}{|z_A|} f(x), \quad (82)$$

where

$$f(x) = \frac{1}{x} \left(\frac{1 - e^{-x}}{1 - ce^{-x}} \right)^2. \quad (83)$$

The two limits are automatic:

$$r \rightarrow 0 \ (x \rightarrow \infty) : \quad P_{\text{erg}} \propto r \rightarrow 0, \quad (84)$$

$$r \rightarrow \infty \ (x \rightarrow 0) : \quad P_{\text{erg}} \propto \frac{1}{r} \rightarrow 0. \quad (85)$$

Thus a finite optimum exists: too slow wastes regrown coherence; too fast collides with a depleted system (Zeno-like freeze-out of extraction).

12.2 Stationarity condition and impedance matching

The condition $d \ln f / dx = 0$ gives

$$2xe^{-x}(1 - c) = (1 - e^{-x})(1 - ce^{-x}). \quad (86)$$

Set $\varepsilon = 1 - c = 1 - \cos\theta$. Solving perturbatively yields

$$x_{\text{opt}} = \varepsilon + \frac{\varepsilon^2}{2} - \frac{\varepsilon^4}{4} + O(\varepsilon^5) \quad (\text{no } \varepsilon^3 \text{ term}), \quad (87)$$

hence at leading order

$$r_{\text{opt}} \simeq \frac{\gamma}{1 - \cos\theta}, \quad \Gamma_{\text{ext}} = r(1 - \cos\theta) \simeq \gamma. \quad (88)$$

This is the impedance-matching condition: the extraction rate equals the bath's coherence-regrowth rate.

12.3 System coherence at the optimum

Using the stationarity equation,

$$\left. \frac{C^*}{c_0} \right|_{\text{opt}} = \frac{1}{2 - x_{\text{opt}}} = \frac{1}{2} + O(\varepsilon). \quad (89)$$

The system is held at about half of its free steady coherence at maximum power.

12.4 Harvesting prefactor and plateau

Define $\mathcal{C}(\theta) = \sin^2 \theta f(x_{\text{opt}})$. The expansion gives

$$\mathcal{C}(\theta) = \frac{1}{2} - \frac{\varepsilon^2}{24} - \frac{\varepsilon^3}{24} + O(\varepsilon^4) = \frac{1}{2} - \frac{\theta^4}{96} + O(\theta^6). \quad (90)$$

The plateau is very flat: the first correction is fourth order in θ . The small-angle harvesting bound is therefore

$$P_{\text{erg}}^{\text{max}} \simeq \frac{\gamma \Delta c_0^2}{2|z_A|}. \quad (91)$$

For the ideal V-system target $c_0 = s_{\text{ss}}/2$ with $s_{\text{ss}} = 1/(1 + e^{\beta\omega})$,

$$P_{\text{erg}}^{\text{max}} \simeq \frac{\gamma \Delta}{8|z_A|} \frac{1}{(1 + e^{\beta\omega})^2}. \quad (92)$$

13 Charging power with finite precession phase

13.1 Exact weak-extraction power function

Keep the resonant finite-gap phase $\phi = \Delta\tau_c = \Delta/r$. Let

$$\delta = \frac{\Delta}{\gamma_C}, \quad x = \frac{\gamma_C}{r}, \quad \phi = \delta x. \quad (93)$$

Then

$$|C^*|^2 = (1 - e^{-x})^2 c_0^2 \left[1 + e^{-2x} \cos^2 \theta - 2e^{-x} \cos \theta \cos(\delta x) \right]^{-1}, \quad (94)$$

and

$$P_{\text{erg}} \propto \frac{1}{x} \frac{(1 - e^{-x})^2}{1 + e^{-2x} \cos^2 \theta - 2e^{-x} \cos \theta \cos(\delta x)}. \quad (95)$$

13.2 Small-angle finite- δ optimum

Let $\varepsilon = 1 - \cos \theta \ll 1$ and assume $x = O(\varepsilon)$. To leading order,

$$1 + e^{-2x} c^2 - 2e^{-x} c \cos(\delta x) \simeq (x + \varepsilon)^2 + \delta^2 x^2, \quad (96)$$

so the reduced objective is $f_\delta(x) \simeq x/[(x + \varepsilon)^2 + \delta^2 x^2]$. Stationarity gives $\varepsilon^2 - (1 + \delta^2)x^2 = 0$, hence

$$x_{\text{opt}}(\delta) \simeq \frac{\varepsilon}{\sqrt{1 + \delta^2}}, \quad r_{\text{opt}}(\delta) \simeq \frac{\gamma_C \sqrt{1 + \delta^2}}{1 - \cos \theta}, \quad (97)$$

and

$$\mathcal{C}_\delta(\theta) \simeq \frac{1}{1 + \sqrt{1 + \delta^2}}, \quad P_{\text{erg}}^{\text{max}}(\delta) \simeq \frac{\gamma_C \Delta c_0^2}{|z_A|} \frac{1}{1 + \sqrt{1 + \delta^2}}. \quad (98)$$

For $\delta = 0$ this recovers $1/2$; for $\delta \gg 1$ the bound is suppressed as $1/\delta$.

Interpretation. The finite gap is needed for nonzero ergotropy, but the same gap causes phase precession between collisions. The best regime is a small but finite matched gap: large enough to define useful battery energy, small enough that $\delta = \Delta/\gamma_C$ does not strongly suppress the coherent harvesting channel.

14 Finite coupling-strength realizability

The formal maximum uses $\theta \rightarrow 0$ and $r \rightarrow \infty$ while keeping $r(1 - \cos \theta)$ fixed. This limit is not automatically realizable because $\theta = g\tau$ and $\tau \leq T = 1/r$, so $\theta \leq g/r$. At the zero-phase optimum $r \simeq 2\gamma_C/\theta^2$, which implies

$$\theta \gtrsim \frac{2\gamma_C}{g}. \quad (99)$$

With finite precession, $\theta \gtrsim 2\gamma_C\sqrt{1 + \delta^2}/g$.

This is the correct *linear* constraint in γ_C/g , not a square-root constraint. Because the zero-phase plateau is flat as θ^4 , moderate g/γ_C can still get very close to the formal bound.

15 Accessibility of coherent ergotropy

15.1 Single ancilla

At $\alpha = 0$ and $z_A < 0$, the output ergotropy is coherent ergotropy. Without a phase reference, a single qubit's coherence between distinct energy levels is not operationally extractable by time-translation-covariant operations [2]. One must distinguish

$$\mathcal{W}_{\text{total}}, \quad \mathcal{W}_{\text{coh}}, \quad \mathcal{W}_{\text{acc}}. \quad (100)$$

For one outgoing ancilla without a reference,

$$\mathcal{W}_{\text{acc}}^{(1)} = 0 \quad (101)$$

in the purely coherent, ground-biased operating regime.

15.2 Two-copy calculation: exact threshold

Let a single outgoing ancilla be

$$\rho_A = \begin{pmatrix} a & c \\ c^* & b \end{pmatrix}, \quad a + b = 1, \quad a > b, \quad (102)$$

with Hamiltonian $H_A = \Delta |\uparrow\rangle\langle\uparrow|$. After global energy dephasing of $\rho_A^{\otimes 2}$, the one-excitation block has eigenvalues $\lambda_{\pm} = ab \pm |c|^2$, while the other eigenvalues are $\lambda_0 = a^2$ and $\lambda_2 = b^2$. For a ground-biased state, two-copy activation requires

$$b^2 > ab - |c|^2. \quad (103)$$

$$\mathcal{W}_{\text{acc}}^{(2)} = \Delta \max \{0, |c|^2 - b(a - b)\}. \quad (104)$$

In the weak ground-ancilla harvesting regime,

$$b \simeq \frac{\sin^2 \theta}{2}(s^* + d^*), \quad |c|^2 \simeq \sin^2 \theta |C^*|^2, \quad (105)$$

so activation needs roughly $|C^*|^2 \gtrsim (s^* + d^*)/2$. Positivity bounds $|C^*|^2 \leq \rho_{11}\rho_{22} \lesssim (s^*)^2/4$, and this condition is generally not met for $s^* \leq 1$ in the weak regime. Two copies usually do *not* unlock the single-copy coherent ergotropy.

Honest statement for a paper. Present total ergotropy as an upper bound. Report one-copy and two-copy accessible pieces. Treat finite- N tape accessibility (possibly with system-mediated correlations) as open unless solved, or include an explicit phase reference and state it clearly.

16 Summary of main analytic results

Result	Formula / statement
Work-free collision	Finite matched gap $\Delta_A = \Delta$ restores $[H_S + H_A, H_{\text{int}}] = 0$ with nonzero battery energy.
Collision map	$s' = s, d' = \cos^2 \theta d - \sin^2 \theta (1 - 2\alpha)s, C' = \cos \theta C.$
Ancilla coherence	$\rho_{\uparrow\downarrow}^{A,\text{out}} = -i \sin \theta (1 - 2\alpha)C.$
Fixed-point coherence	$C^* = (1 - q_C)c_0 / (1 - q_C e^{-i\phi} \cos \theta).$
Directed leakage FP	$v^* = (\mathbb{K} - \mathcal{B}R\theta)^{-1}b_{\mathcal{B}}.$
Qubit ergotropy	$\mathcal{W}_A = \Delta(z_A + r_A)/2.$
Impedance matching	$\Gamma_{\text{ext}} = r(1 - \cos \theta) \simeq \gamma_C.$
Zero-phase bound	$P_{\text{erg}}^{\text{max}} \simeq \gamma_C \Delta c_0^2 / (2 z_A).$
Finite-phase bound	$P_{\text{erg}}^{\text{max}}(\delta) \simeq \gamma_C \Delta c_0^2 / z_A \cdot 1/(1 + \sqrt{1 + \delta^2}).$
Realizability	$\theta \gtrsim 2\gamma_C \sqrt{1 + \delta^2} / g.$
Plateau	$\mathcal{C}(\theta) = 1/2 - \theta^4/96 + O(\theta^6).$
Two-copy access	$\mathcal{W}_{\text{acc}}^{(2)} = \Delta \max\{0, c ^2 - b(a - b)\}$ (often zero in the weak regime).

17 Observables, figures, and next calculations

17.1 Recommended figures

1. Diagram: bath pumps bright state; collisions drain bright–dark imbalance to the ancilla tape.
2. Collision-map validation: analytic versus numerical partial trace on the 6×6 joint space.
3. Fixed point: $|C^*|/c_0$ versus r/γ_C for several θ .
4. Power curve: $P_{\text{erg}}(r)$ showing slow-collision and Zeno limits.
5. Impedance matching: optimum at $\Gamma_{\text{ext}}/\gamma_C \simeq 1$.
6. Plateau: $\mathcal{C}(\theta)$ versus θ , highlighting the flat small-angle region.
7. Finite gap: suppression versus $\delta = \Delta/\gamma_C$.
8. Directed leakage: resource suppression versus κ_D/γ_C .
9. Accessibility: total ergotropy versus one- and two-copy accessible work.

17.2 Immediate calculations still needed before a short paper

1. Implement the directed dark-leakage fixed point and compare it to the phenomenological reset map.
2. Plot exact $P_{\text{erg}}(r)$ from the exact ergotropy formula, not only the weak expansion.
3. Numerically verify the finite-phase optimum and the $1/(1 + \sqrt{1 + \delta^2})$ law.
4. Check whether any realistic parameter set activates two-copy accessible work.
5. Decide and state the phase-reference assumption, or derive the actual near-degenerate V-system Liouvillian.
6. Optional but high value: estimate preparation free-energy cost ΔF_A and report a net figure of merit.

17.3 Only after the companion model is closed

Move to the full spin-valve engine with

$$\rho_D^{(n+1)} = \text{Tr}_A \left\{ U_\tau \left[e^{\mathcal{L}_{\text{leads}}(T-\tau)} \rho_D^{(n)} \otimes \rho_A^{\text{in}} \right] U_\tau^\dagger \right\}, \quad (106)$$

using a nonsecular lead generator. The first full-engine deliverable should be the tradeoff curve $(P_{\text{el}}, P_{\text{erg}})$, plus controls that distinguish population resetting from genuine coherence harvesting—not a claim of restored collinear power.

18 Open decisions and risk points

Issue	Why it matters	Recommended treatment
Finite-gap coherent bath	Precession vs. stationary lab-frame coherence.	Phase-locked companion model, or $\Delta \ll \gamma_C$, or derive the actual V-system Liouvillian.
Dark-state renewal	Collisions can trap population in the dark state.	Directed dark leakage or weak symmetry breaking.
Accessible work	Single-copy coherent ergotropy is not free.	Upper bound + finite- N or explicit reference.
Preparation cost	Ground-state ancillas are a low-entropy resource.	Gross ergotropy first; later ΔF_A .
Full transport generator	Secular Lindblad kills spin coherence.	Companion first; then nonsecular leads with positivity checks.
Narrative risk	Harvesting may not improve electrical power.	Frame as resource partitioning, not guaranteed restoration.

19 Outlook

The spin valve remains the physically rich target: it supplies a continuously regenerated, transport-driven coherent resource and a second output channel (electrical power) against which harvesting can be traded. The companion model fixes the harvesting physics cleanly: collision algebra, finite-gap repair, impedance matching, and the leading finite-phase suppression law. Once the bath mechanism and accessibility bookkeeping are locked, the companion model is strong enough to stand as a short analytic paper or as the analytic core of a longer transport paper.

On the experimental side, the exchange interaction (9) has a natural double-dot realization. Coherent spin-swap control requires low-temperature platforms rather than the ~ 100 K scale of the legacy parameters; a feasibility discussion must specify g , γ , τ , T , spin-dephasing times, and ancilla initialization and readout.

Bottom line. The collision algebra is sound, the finite-gap repair makes a work-free battery interpretation possible, and impedance matching is a strong organizing principle. The two remaining integrity checks are (i) a directed (not symmetric) renewal bath and (ii) honest accessible-work accounting. With those closed, the companion model is a real analytic result; the spin-valve engine is then the natural extension.

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